TKG-RAG: A Retrieval-Augmented Generation Framework with Text-chunk Knowledge Graph

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***Abstract*—The approach of dividing text into chunks and building indexes is a mainstream approach for Retrieval-Augmented Generation (RAG) of Large Language Models (LLMs). However, retrieved text chunks often contain noise or redundant information, which can negatively impact RAG performance. To address this limitation, researchers have proposed RAG approaches based on text chunks, knowledge graphs, and long-texts. Nevertheless, there are still challenges that need to be addressed, such as the underutilization of knowledge within text chunks, the resource-intensive nature and complex process of constructing Knowledge Graphs (KG), and the lack of emphasis on optimization during the post-processing filtering stage. We propose a RAG framework with text-chunk knowledge graph (TKG-RAG). The proposed framework can construct a text-chunk knowledge graph automatically by extracting the hierarchical structure, contextual relationships, topic sentences, and inter-chunk relationships from the domain text. The framework begins by using text indexing to retrieve relevant text chunks based on similarity. These chunks are then mapped to nodes within the text-chunk knowledge graph, where connections are established based on the relationships of nodes in the graph to generate subgraphs. The content of the nodes is rearranged and merged using the hierarchical structure and relational knowledge in the graph, and then the residual isolated nodes that are not merged are fused based on the attribute knowledge in the TKG. In addition, to improve the performance of filters in the post-processing stage, we incorporate the datasets considering texts and numerical characteristics, into the fine-tuning process of the filter model. This approach aimed to enhance the accuracy of filtering and reduce token consumption in the generation stage. To validate the effectiveness of this approach, comparative and ablation experiments were conducted on five datasets: NQ, PopQA, HotpotQA, TriviaQA, and LawQA. The results show that TKG-RAG can achieve better performance in terms of Accuracy and F1 scores, while also reducing token consumption by 46%, means that TKG-RAG can combine the strengths of chunk-based and graph-based RAG approaches.**

Keywords—retrieval-augmented generation, information mining, multidimensional knowledge, text chunk knowledge graph, fusion enhancement

I. Introduction

To address the issue of outdated knowledge in Large Language Models (LLMs), Lewis et al. (2020)[1]introduced the Retrieval-Augmented Generation (RAG) method, which enhances models by incorporating external knowledge. However, empirical studies have shown that improper information fusion within RAG may introduce noise, which deteriorates RAG performance[2]. Consequently, researchers have focused on optimizing information fusion process to enhance the effectiveness of models in downstream tasks.

Current research on fusion methods falls into three main approaches: text-chunk based, knowledge graph based, and long-text based RAG. Among these, the text-chunk based approach is widely adopted due to its simplicity and ease of implementation. However, it often fails to capture inter-chunk relationships, leading to issues such as error propagation and information loss, thereby limiting the overall effectiveness of RAG[3][4]. The knowledge graph-based approach provides more comprehensive and accurate responses, but it is computationally and structurally complex, less efficient for detailed tasks, and requires significant computational resources[5]. The long-text based approach eliminates the need for complex chunking strategies and knowledge graph extraction, but it has some limitations on the model due to its need a long context window. Additionally, in scenarios involving long-text responses, this approach may suffer from information loss[6][7].

After the fusion process, content filtering is necessary to enhance content quality. Existing research typically uses fine-tuned filter models to score content and exclude low-scoring entries. However, when constructing the training data for the filters, corpus diversity is often neglected, particularly the inclusion of texts containing numerical data, leading to reduced filtering accuracy[2][8].

To solve the shortcomings of current RAG approaches, we proposes a RAG framework with text chunk knowledge graph. The framework can automatically construct a text chunk knowledge graph by extracting embedded knowledge and then leveraging the knowledge in the KG for information fusion. Finally, a filter is employed to extract content relevant to the query, minimizing token consumption and improve processing efficiency. The framework not only makes use of additional information like the hierarchical structure of the KG but also reduces the complexity involved in constructing the KG.

The main contributions of this paper include the following three aspects:

1) Proposing a Text-chunk Knowledge Graph: This work introduces a text chunk knowledge graph and leverages it to optimize RAG. It uses both attribute knowledge and relational knowledge within the KG for effective information fusion.

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2) Fine-tuned Filtering for Different Types of Corpus: During the filtering stage, a filter fine-tuned with diverse types of corpus is employed to filter the noisy information. The filter model can accurate assessment of content relevance to the query, preventing error propagation.

3) Empirical Validation through Comparative and Ablation Experiments: The performance improvements of TKG-RAG are demonstrated through comparative and ablation experiments across five question-answering datasets: Natural Questions[9], PopQA[10], TriviaQA[11], HotpotQA[12], and LawQA. The results demonstrate a 4.38% average accuracy improvement for the generated answers while reducing token consumption in the generation stage by 46%.

II. Related Work

This section first introduces the existing RAG fusion approaches, focusing on three approaches: text-chunk based, knowledge graph based, and long-text based, along with the subsequent filter process. It details the implementation of representative approaches, analyzes their advantages and disadvantages, which deserve more in-depth research.

A. RAG Fusion Method

The RAG fusion approaches can be broadly categorized into three types, with a detailed analysis provided in TABLE I.

The first category is text-chunk based approaches, which can be further divided into three subtypes: summarizing chunks, using generated metadata to guide LLMs in query rewriting, and utilizing the hierarchical structure of text for retrieval. Regarding summarizing chunks, Sarthi et al.[3] proposed a recursive approach for generating hierarchical summaries, constructing a tree structure that clusters similar chunks into summary nodes at various abstraction levels, thereby enhancing retrieval performance. However, due to the recursive nature of summary generation, there is a risk of error propagation or loss of details, which can negatively impact the performance when answering detailed questions. Then, regarding the metadata query rewriting approach, Mombaerts et al.[13] generated metadata for each document, including summaries and question-answer pairs. Upon receiving a query, the system first retrieves the metadata summary and combines it with user profiles to perform query rewriting before proceeding with retrieval and reading stages. This approach improves the quality of the final answer by leveraging metadata in the query rewriting phase. Finally, regarding the use of hierarchical structure for retrieval, Chen et al.[4] used LLMs to parse unstructured documents into markdown format, extracting hierarchical structure information and cascading metadata into each chapter. They applied a multi-level retrieval method combining vector similarity, keyword matching, and BM25 to enhance RAG. By integrating these different matching techniques, this approach effectively utilizes the structural information within documents to improve retrieval results.

The second category is knowledge graph based approaches, which can be divided into two subcategories: using existing KG and constructing new one. The first subcategory involves using existing KG. For example, Yu et al.[14] utilize a mapping function to link paragraphs to text entities within a KG. They apply a two-stage re-ranking process using Dense Passage Retrieval (DPR) and graph neural networks (GNN), aiming to refine the retrieval results. It was later found that existing KG were insufficient for adapting to a variety of tasks, leading to the construction of specialized and precise KG. For instance, Edge et al.[5] construct a graph-based text index during the preprocessing stage to create an entity knowledge graph and generate community summaries for groups of entities. During the inference stage, the final answer is derived by summarizing responses from multiple communities. This approach performs well in query-centered summarization tasks but is less effective when dealing with specific details. Additionally, the approach is resource-intensive, with high demands on model performance and token consumption in both the preprocessing and inference stages. Xu et al.[15] significantly improve the accuracy of retrieval and answer generation by constructing a KG from historical customer service queries. This approach allows for a more precise retrieval process and enhances the quality of generated responses by leveraging past interactions.

The third category is long-text based approaches. For example, Jiang et al.[6] fine-tuned a Compressor using data distillation to compress the context, optimizing the use of long texts for retrieval. This approach aims to reduce the size of the input while retaining the most relevant information, enhancing the ability of model to handle longer contexts efficiently. Unlike the model fine-tuning approach, Liu et al.[7] introduced a retrieval-memory module to store past contextual knowledge. When memory overflow occurs, the system dynamically retains valuable information and discards irrelevant data. This approach extends the context length from 4k to 80k tokens, allowing the model to process and utilize much longer contexts during retrieval and generation tasks.

TABLE I. Analysis of advantages and disadvantages of RAG related work

Type of Approach	Advantage	Disadvantage	Method
text-chunk based approach	Clear hierarchy, Structured processing, Multiple retrieval, Metadata driven, Personalized query	Erroneous propagation, Loss of detail information, Underutilization of chunk information, Recall bias	RAPTOR[3], HiQA[4], MK-RAG[9]
knowledge graph based approach	Global perspective, Community summaries, Structured relationship, Problem structure retention	Complex and resource-intensive graph construction, High difficulty in graph retrieval	Graph-RAG[5], KG-FiD[14], KG-CSQA[15]
long-text based approach	Semantic integrity, High resource utilization	Bias in the fine-tuned compression model dataset, Loss of precision in compression	LLMLingua[6], MemLong[7]
filter after fusion approach	Robustness, Adaptive correction, Multi-source information fusion	Insufficient consideration of potential text types, Insensitivity to numerical features, leading to error loss	CRAG[2]、FILCO[8]、RE-RAG[16]

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B. RAG Filter Method

Since the fused information may contain noise, researchers apply filtering techniques to improve the signal-to-noise ratio, thereby enhancing the performance of the RAG system. Current approaches typically achieve this by fine-tuning a filter. For instance, Wang et al.[8] employed diverse methods to construct datasets requisite for fine-tuning filters, subsequently utilizing the fine-tuned model to estimate contextual relevance based on conditional cross-mutual information. Then, Kim et al.[16] fine-tuned a scoring model to estimate the relevance of retrieved documents, followed by a refinement process to effectively filter irrelevant knowledge. In particular, Yan et al.[2] built upon conventional scoring systems, focusing on the use of the external knowledge to enhance model performance in cases where retrieval results are inaccurate or lack answers. In the post-processing phase, where the relevance of retrieved information chunks is assessed for filtering, their approach considered only the relevance between the query and individual chunks. This overlooked the explicit and implicit relationships between chunks, resulting in underutilization of internal information and, in some cases, the unintended removal of valuable information.

III. Approach

This section introduces the RAG framework based on a text-chunk knowledge graph, referred to as the Text-chunk Knowledge Graph RAG (TKG-RAG). It provides a detailed analysis of the design and implementation of the text-chunk knowledge graph constructor and the information fusion enhancer. Furthermore, it explains how the knowledge embedded within the TKG is utilized to improve the overall performance.

A. Task Formulation

Given a question q and a document set $D = \{d_1, d_2, \cdots d_n\}$, Information such as id, title, topic sentence is extracted from d_i to construct a text-chunk knowledge graph $\mathcal{G}$. The nodes of the TKG include text chunks and document files, with relationships representing contextual connections between chunks or the containment relationships between files and chunks. Through a retriever, k text chunks $C_k^q = \{c_1^q, c_2^q, \cdots, c_k^q\}$ relevant to the question q are retrieved from document set D based on similarity.

Each chunk c_i^q is mapped to its corresponding position in $\mathcal{G}$ through a mapping function $f(c_i^q)$ using its id. This process recalls a corresponding subgraph g, which leverages the structured knowledge within the graph for fusion and enhancement. After fusion and filtering (FR), a set $FR = \{fr_1, fr_2, \cdots, fr_m\}$ of m subsets is obtained. This set is then concatenated with the question and fed into a LLM to generate the answer, $y = \{LLM(p, q, FR)\}$, where p represents the prompt and y is the response of LLM.

B. Overall Framework

The proposed TKG-RAG leverages the knowledge of individual chunks and the relationships between them to perform relevance analysis, enabling the fusion and filtering of context, thereby reducing information noise and redundancy. The overall framework of TKG-RAG is illustrated in Fig. 1. TKG-RAG consists of four components: (1) Text-chunk Knowledge Graph Constructor, which extracts knowledge of chunks from the corpus to build a text-chunk knowledge graph in the RAG framework; (2) Text Retriever, which retrieves candidate chunks from the corpus and maps them to the TKG for graph-based retrieval; (3) Information Fusion Enhancer, which fusions and enhances both structured and unstructured knowledge retrieved from the graph and filtering; (4) Question Answer Generator, which combines the question with the filtered information to predict the final answer.

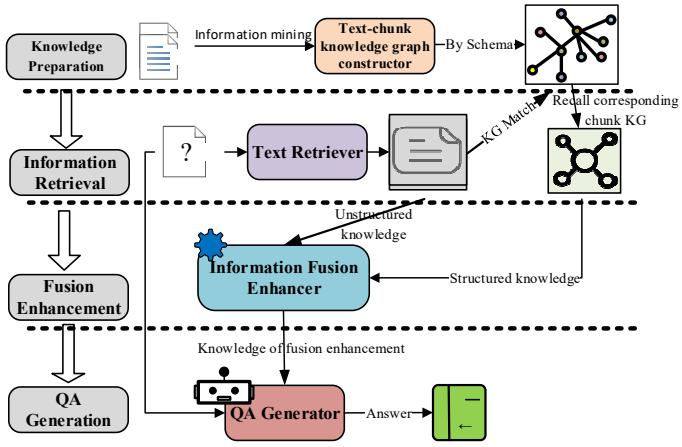

Fig. 1. TKG-RAG Overall Framework.

1) In the knowledge preparation phase, a text-chunk knowledge graph is automatically constructed and stored in a graph database by extracting the hierarchical structure, contextual relationships, topic sentences, and inter-chunk relationships, based on a predefined schema.

2) In the information retrieval phase, the process begins with vector similarity-based text retrieval based on the user input question, resulting in a series of text chunks as unstructured knowledge. These text chunks are then used to recall corresponding subgraphs from TKG by a mapping function, providing structured knowledge.

3) In the fusion and enhancement phase, the unstructured knowledge from text retrieval is combined with the structured knowledge from graph retrieval. This process initiates with the fusion of knowledge through the hierarchical structure and interrelationships among chunk nodes in TKG; subsequently, further fusion occurs based on the attribute knowledge of the nodes in TKG. Finally, a scoring mechanism is implemented to evaluate and select the most relevant knowledge.

4) The final answer is generated by the question-answer generator, which combines the user input question with the knowledge filtered through fusion.

Benefiting from the Text-chunk Knowledge Graph Constructor and the Information Fusion Enhancer, TKG-RAG can effectively provide enhanced inputs for LLM. The following sections will provide a detailed analysis and description of these components.

C. Text-chunk Knowledge Graph Construction

The rationale for using a knowledge graph to represent information chunks in RAG lies in its ability to capture both explicit and implicit relationships between chunks[14]. In addition, the attribute knowledge in TKG can be dynamically

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modified or enriched, which is beneficial for improving performance during the fusion and enhancement phase.

The process begins with ontology modeling, which defines a set of relationships among conceptual entities such as file, chapter, and chunk. Additionally, the entities include various attributes, as shown in the schema of the text chunk knowledge graph in Fig. 2. Rule-based methods are used to extract attribute knowledge such as the id, title, hierarchy of each chunk, while the topic sentence is derived using LLM. The TKG is automatically constructed by treating chunks, chapters, and files as nodes, while utilizing hierarchical and containment relationships as edges.

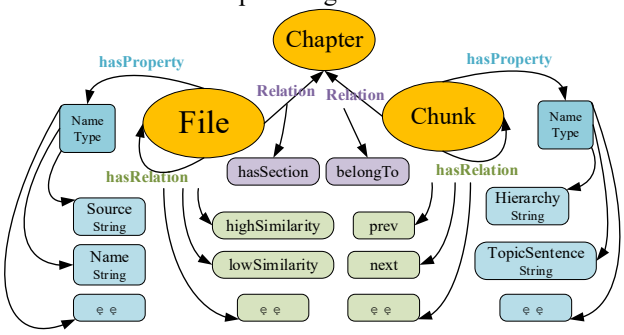

Fig. 2. The schema of Text-chunk Knowledge Graph.

In the Wikipedia text corpus, each article is divided into several non-overlapping text chunks, containing 100 words, which serve as the basic retrieval unit. This approach raises the following issues: (1) The segmentation process divides the text directly into chunks of 100 words, which may result in sentence fragmentation, leading to incomplete semantic units. (2) A subject that appears in the first half of a paragraph may be replaced by a pronoun upon its subsequent mention. As a result, in the segmented text, the latter part of the sentence consists solely of the pronoun, failing to convey the actual subject and resulting in incomplete information.

Therefore, this study concatenates the title of the chunk with the corresponding actual text inputs to the LLM, prompting it to generate topic sentences for each text chunk. This approach of using topic sentences can guide the LLM in analyzing the context, enabling it to restore pronouns and include complete information. By leveraging the capability of model, it allows for the extraction of underlying knowledge within the text chunks. For example, in chunk c_5 in Fig. 3, the LLM can identify information like "There were four wars between the two countries.".

TABLE II. displays the prompts used to generate these topic sentences, guiding the LLM to replace pronouns with their full references, thereby preserving the original meaning of the sentences to the greatest extent.

TABLE II. PROMPT FOR GENERATING TOPIC SENTENCES

Given the following text segment, which include the title and body content, generate a series of concise summaries and make up the final summary list, which encapsulates the main topics discussed in the text. The summary should be in sentence form and reflect the main points of the article. In each summary, if there is a pronoun, restore it to the full word it refers. Each topic is output with a newline, without any special markup sort symbols. There is no need to output any additional content. Title: {title}, Content: {context}.

D. Information Fusion Enhancement

Most retrieved chunks lack organization and structure. From a human behavior perspective, clustering the retrieved results and then analyzing them individually is an effective strategy. This framework leverages the knowledge within TKG for fusion and enhancement, dividing the retrieved chunks into multiple sets and ranking them internally. This organization of clusters facilitates subsequent filtering and answer generation. Additionally, relevant studies[17] highlight that clustering is beneficial for document consolidation, improving retrieval and understanding. By exploring the internal group structure of input data and providing high-quality feature representations, the performance of downstream tasks can be significantly enhanced.

Fusion The fusion enhancement consists of two subcomponents that leverage different types of knowledge: one leverages attribute knowledge from the TKG, while the other leverages relational knowledge from the TKG. Ablation experiments were conducted in the experimental section to determine the differences and performance gains associated with using different types of knowledge. The approach for fusion enhancement using relational knowledge from TKG expands the information beyond the limitations of vector retrieval results, making it more comprehensive and enriching. For a recalled chunk node denoted as c_i^q, we identify its location in the TKG and initially explore outward to nodes at a distance of *step*. Then, we leverage relationships such as "*belongTo*" and "*hasSection*" to trace upwards to the chapter node. From this chapter node, we further explore horizontally to find chapter nodes that exhibit a "*highSimilarity*" relationship. Next, we dig downwards to uncover nodes related to the initial node c_i^q and the question q. Subsequently, for the recalled subgraph, we disregard the influence of similarity scores obtained from text retrieval. Based on the relationships "*prev*" and "*next*" within the subgraph, as illustrated in the information fusion module on the right side of Fig. 3, we connect adjacent text chunk nodes in their original semantic order. This allows for the rearranging of the recalled chunks c_i^q, adding those that meet these conditions into a new document set FR.

After performing node rearranging fusion with the TKG, there are still some isolated nodes that have not been merged. These nodes originate from various documents, have varying hierarchical structures, and lack connections through the relationships of graph. Despite differences, these nodes convey similar meanings. This framework employs the agglomerative hierarchical clustering algorithm[18], which organizes data points into a hierarchical structure without requiring a predefined number of clusters. As shown in the information fusion module in Fig. 3, this approach incorporates the attribute knowledge of the nodes in the TKG, such as title and topic sentences, into the original text content, as illustrated in Algorithm 1.

Algorithm1: KG-based RAG Fusion

Input: retrieved_chunks(top-k retrieved_chunks), TKG(Text-chunk Knowledge Graph)
Output: final_clusters (Fused knowledge clusters)
mapping_kg ← Map retrieved_chunks to TKG nodes
for each node n in mapping_kg do
  neighbors ← ExploreNeighbors(n, step)
  recall_kg ← recall_kg ∪ neighbors
  section_nodes ← TraceUpwards(n, relations)

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```
    recall_kg ← recall_kg ∪ section_nodes
end for
stage1_cluster ← SortChunks(recall_kg, relations)
remaining_kg ← recall_kg \ stage1_cluster
for each node n in remaining_kg do
    n.topic_sentence ← ExtractTopicSentence(n)
end for
// AHC: Agglomerative Hierarchical Clustering Algorithm
stage2_cluster ← AHC(remaining_kg)
final_clusters ← stage1_cluster ∪ stage2_cluster
```

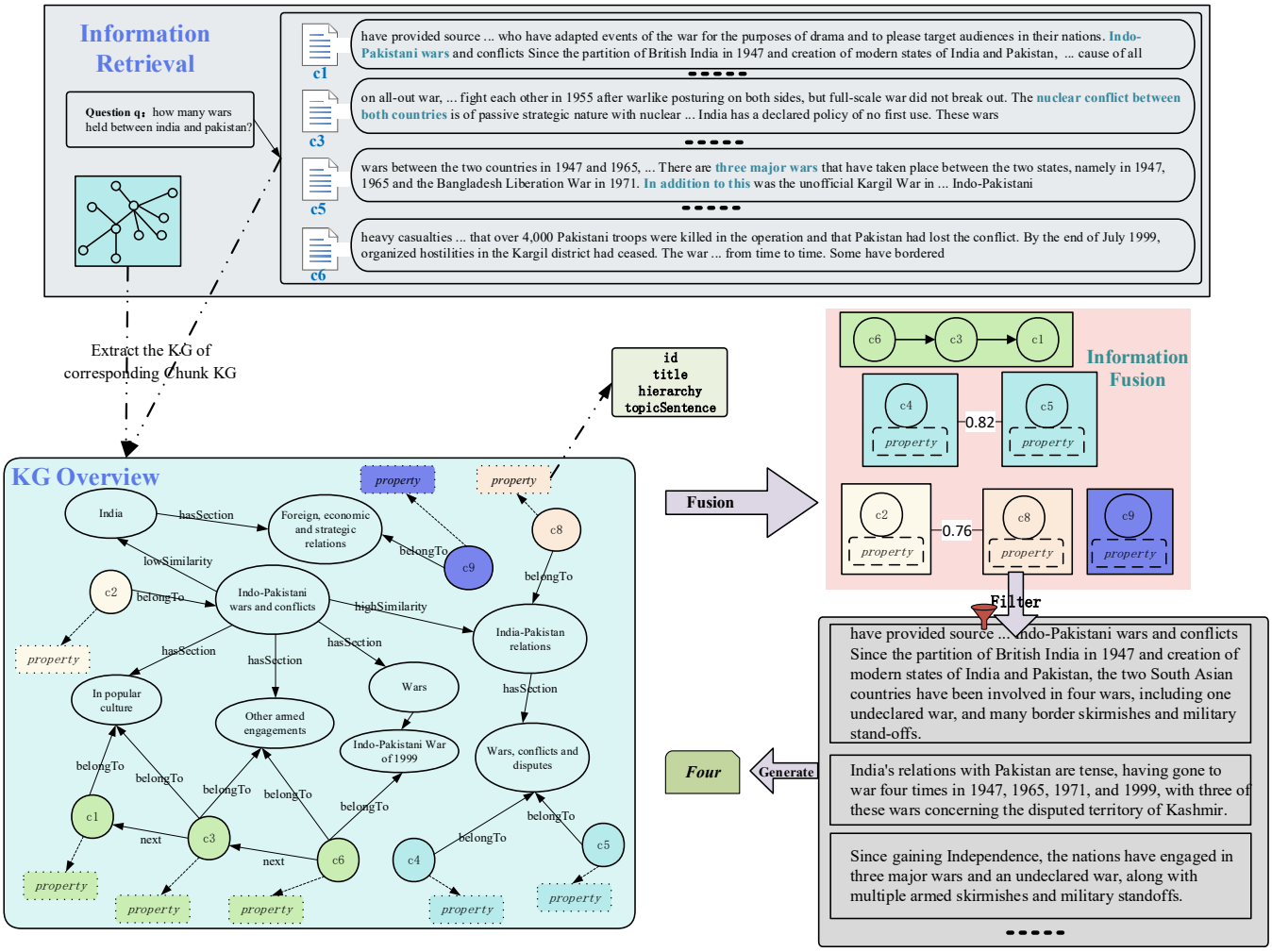

Fig. 3. Detailed diagram of the TKG-RAG process.

Compared to using vector retrieval for RAG, the introduction of a text-chunk knowledge graph enhances the utilization of information through attribute knowledge and enriches the results of information retrieval through relational knowledge from TKG. In contrast to RAG approaches that solely utilize knowledge graphs, TKG-RAG emphasizes the implicit and explicit knowledge of the chunks themselves as well as the hierarchical and contextual relationships between chunks. This framework eliminates the need for an LLM to extract entities and relevant triples, thereby reducing the complexity and costs associated with their construction. Thus, this approach balances the efficiency of vector retrieval with the structured representation of knowledge graphs.

In-depth analysis reveals that the set FR, which comprises the information filtered through the fusion process, contains attribute and relational knowledge related to chunk c_i^q and the expected answer y. This information originates neither from external knowledge retrieval nor from independent model generation but is derived from existing data within the system. Therefore, the introduction of FR does not introduce additional randomness; instead, it contributes to reduce the entropy of the RAG system.

In traditional RAG systems, the method solely utilizes the question and the retrieved text, represented by the formula $p(y|q,C)$, which denotes the probability of the model generating the answer y given the question q and the context C. The conditional entropy is expressed as shown in Equation (1):

$$H(Y \mid q,C) = -\sum_{y} p(y \mid q,c) \log p(y \mid q,c) \tag{1}$$

The conditional entropy of the proposed text-chunk knowledge graph approach is represented by Equation (2):

$$H(Y \mid q,C,FR) = -\sum_{y} p(y \mid q,c,fr) \log p(y \mid q,c,fr) \tag{2}$$

According to the chain rule of conditional entropy, as expressed in Equation (3):

$$H(Y \mid q,C,FR) = H(Y \mid q,C) - I(Y;FR \mid q,C) \tag{3}$$

$I(Y;FR|q,C)$ denotes the mutual information between Y and FR given q and C. Since mutual information is always

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non-negative, we have $I(Y; FR|q, C) \geq 0$. Consequently, we can derive Equation (4):

$$H(Y \mid q, C, FR) \leq H(Y \mid q, C) \tag{4}$$

This indicates that fusing additional information g to obtain set FR can reduce or, at the very least, not increase conditional entropy. Consequently, this approach has the potential to improve the accuracy of RAG systems. Moreover, recent research by Zhu et al.[19] has employed mutual information to calculate the information bottleneck, integrating this concept into RAG systems to optimize noise filtering. This metric was subsequently applied in reinforcement learning to further improve the performance of RAG systems.

When utilizing metadata such as hierarchical structures prior to retrieval, the RAG system can only recall chunks directly related to the query, which constrains the range of retrieval results and may lead to the loss of certain details or indirectly relevant yet valuable information. However, employing metadata after retrieval allows for a more comprehensive exploration and backtracking within the text-chunk knowledge graph. By leveraging the hierarchical structure, the system can further retrieve other chunks located under the same subsection as the original chunk, identified by its chapter and subtitle.

This approach effectively facilitates both "lateral expansion", which involves seeking related information from similar contexts, and "vertical backtracking", which entails revisiting and deepening the understanding of the initial retrieval results, thereby enabling the acquisition of broader contextual information, including diverse perspectives and deeper insights, while preserving the relevance of the initial results. This methodology is particularly suited for addressing questions that require background knowledge, such as the question illustrated in Fig. 3: " how many wars held between india and pakistan?" If this information is utilized prior to retrieval, the search scope is limited to the document titled "India and Pakistan." Conversely, if this information is applied after retrieval, it can trace back to a wealth of knowledge, including information sourced from the "India and Pakistan" title as well as related information from other titles, such as "India-Pakistan relations," which provides additional relevant details.

Moreover, during the retrieval phase, similarity analysis is conducted on a one-to-one basis between the question and the chunk. However, after the retrieval, multiple chunks and their associated attribute knowledge can be utilized, allowing for a richer set of information to be compared and analyzed from multiple perspectives.

Filter After information fusion, a fine-tuned $filter$ evaluates the information, applying a predefined threshold for content filtering.

The $filter$ receives the question q and the $text$, producing binary outputs of 0 or 1. These outputs are then normalized to obtain the final relevance metric, allowing the probabilities to be converted into logits that serve as relevance scores.

$$FR_i = \left\{ filter\left(q, text_i\right) \right\}_{i=1}^{m} \tag{5}$$

Regarding the filter component, this study differentiates itself from other researchers by incorporating the context of the chunks during the scoring process. Currently, most approaches evaluate a single question and a retrieved text chunk independently, which can lead to the fragmentation of chunks, resulting in incomplete information and lower scores for both split chunks. In contrast, TKG-RAG scores the results of a cluster enhanced through the previous fusion step, thereby restoring the relevant chunks and ensuring that the semantic information is complete. Furthermore, even if the intended outcome of information restoration is not achieved, the filtering process leverages not only the content of the text chunks but also incorporates the attribute knowledge associated with the corresponding topic sentences derived from TKG. These topic sentences also encapsulate comprehensive content, providing crucial context and insights independently.

In existing RAG frameworks, certain approaches utilize more granular scoring for information chunks[2], often involving sentence lengths that are significantly shorter than those of the original chunks. However, during the fine-tuning of the filter, text length has not been integrated as a feature in the training process. TKG-RAG enhances the training dataset by incorporating both long text chunks and shorter topic sentences. This enables the LLM to effectively capture differences in semantic relationships, thereby improving the comprehensiveness of dataset with respect to text length. Additionally, the relevance assessment also takes into account the sensitivity of numerical information. In cases where the question includes specific years or episodes of a particular television series, this numerical awareness becomes essential.

IV. Experiment

To validate the effectiveness of the proposed TKG-RAG approach, experiments were conducted using four English open-domain question-answering datasets and one Chinese legal domain question-answering dataset. The overall fusion enhancement experiment is presented initially, followed by ablation experiments that evaluate the effectiveness of different types of knowledge employed during the fusion phase, as well as the filters applied during the filtering stage.

A. Experimental Setup

1) Datasets and Evaluation Metrics

To evaluate performance on knowledge-intensive tasks, five datasets were selected: the open-domain question-answering datasets Natural Questions[9], PopQA[10], and TriviaQA[11], the multi-hop question-answering dataset HotpotQA[12], and the legal question-answering dataset LawQA[1], which is derived from regulations established by local emergency management departments. For each dataset, data from the training set was utilized to fine-tune the filters, and 500 samples were randomly selected from the test set to serve as the evaluation dataset. Among these datasets, the F1 score was used as the evaluation metric for LawQA, while the other four datasets employed Accuracy as the evaluation metric.

2) Experimental Details

[1] https://drive.google.com/file/d/1_Z8Ot0eqrwYDLGG0EhKShBHaHaI3RK-S/

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Retrieval Source: The 2018 version of Wikipedia was utilized as the source for the retrieval process, derived from DPR. The corpus consists of 21,015,324 non-overlapping paragraphs, each containing 100 words.

Retrieval Module: The GTR[20] retriever was employed for the retrieval process, incorporating two BERT models designed to encode questions and paragraphs separately.

Topic Sentence Generation Module: The pre-trained model Qwen2-72B-Instruct was used for generation, with the temperature set to 0.7.

Filtering Module: The T5-large model was utilized to initialize the retrieval filter and fine-tune it, with the implementation based on the Hugging Face Transformers library.

Question-Answer Generation Module: The Llama2-7B-Chat and Qwen2-7B-Instruct models served as the backbone models for the first four English datasets and the final Chinese dataset, respectively. Each model has approximately 7 billion parameters and is specifically tailored for dialogue tasks, excelling at understanding and engaging in conversations.

Additional Information: The inference for the above modules was conducted using FastChat[2] and vLLM[21]. All training and evaluation were performed on a single NVIDIA A800 GPU with 80GB of memory, and during the final question-answering inference task, the model employed greedy decoding.

3) Approaches of Comparison

Experiments were conducted on the aforementioned dataset, and the results were compared against the following approaches:

(1) Utilizing the inherent knowledge of model (NoN-RAG): This approach involves directly presenting the question to the model and prompting it to generate an answer. This approach serves to assess the baseline performance of model on the current dataset.

(2) Enhancement through complete chunks (Naïve-RAG): A common approach involves concatenating the question with the retrieved chunks from the retriever, separated by line breaks, and incorporating this into the prompt. This approach serves as the baseline for retrieval-augmented generation.

(3) Advanced RAG: RAPTOR[3] recursively generates summaries and employs multi-level clustering to enhance retrieval accuracy, while Graph-RAG[5] optimizes retrieval accuracy and the completeness of the final output through graph-based text indexing. For resource efficiency, both methods utilize the same model, Qwen2-72B-Instruct, as used in this paper for generating topic sentences during the preprocessing phase.

B. Experimental Result

1) Analysis of Results

The overall performance of different approaches on the five test sets is detailed in TABLE III.

TABLE III. THE OVERALL PERFORMANCE OF DIFFERENT APPROACHES

Approach	NQ	PopQA	TriviaQA	HotpotQA	LawQA
NoN-RAG	32.1	27.3	50.2	22.6	33.8
Naïve-RAG	51.3	55.7	61.4	37.5	58.7
RAPTOR	47.5	36.8	58.2	33.7	48.2
Graph-RAG	50.6	43.3	52.1	**52.0**	44.6
TKG-RAG(ours)	**60.1**	**60.2**	**65.4**	38.2	**62.6**

Compared to the baseline NoN-RAG, which does not utilize a retriever, TKG-RAG shows improvements across each dataset, with an average enhancement of 24.1%. This result underscores the necessity of incorporating external knowledge. When compared to the baseline method Naïve-RAG, TKG-RAG achieves an average accuracy improvement of 4.38%, demonstrating the effectiveness of the augmented fusion approach. In the HotpotQA dataset, the performance improvement of RAG compared to the baseline is less pronounced than in other datasets. This may be attributed to the increased complexity of multi-hop tasks, which differ from extractive tasks.

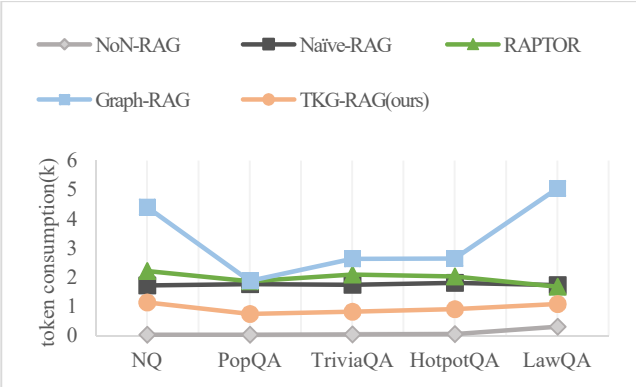

Fig. 4. Token consumption for each approach.

The token consumption of each approach is illustrated in Fig. 4. TKG-RAG achieves an average reduction of 46% in input tokens, indicating that this approach not only enhances the performance of various models on tasks but also improves computational efficiency by eliminating irrelevant content. This demonstrates the significant advantages of augmented fusion, which reduces token consumption while increasing accuracy. Additionally, it is important to note that Graph-RAG requires approximately 24,678,703 tokens to construct graph-based text indices and entity community summaries during the preprocessing phase. Even with vLLM accelerating the LLM reasoning process, it still takes around three days, resulting in significantly higher consumption compared to the other four methods.

Reducing the token count offers considerable benefits in real-world applications, primarily by reducing system computational costs and improving overall processing efficiency, which contributes to an enhanced user experience. However, an excessive focus on processing speed and token reduction can result in significant information loss, which may undermine the overall effectiveness of system.

Therefore, the system must balance token reduction with the retention of essential information. Fig. 5 presents experimental results on four datasets, demonstrating how adjustments in filter threshold values affect token consumption and accuracy. Here, the threshold corresponds to the logits output from the filter, as discussed in the information fusion enhancement section. A higher threshold

[2] https://github.com/lm-sys/FastChat

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results in decreased information retention. Even at extreme threshold values, such as -1 or 0, some content may still be filtered due to the probabilistic output of logits beyond the specified range. In this experiment, for most datasets, accuracy and token consumption are inversely proportional. However, on the HotpotQA dataset, this trend is less evident. This is likely due to the nature of the fine-tuning filters, which are primarily designed to handle simpler question types. In the case of multi-hop questions, the filters may exclude information that is indirectly related to the questions, leading to a weaker correlation between accuracy and token consumption.

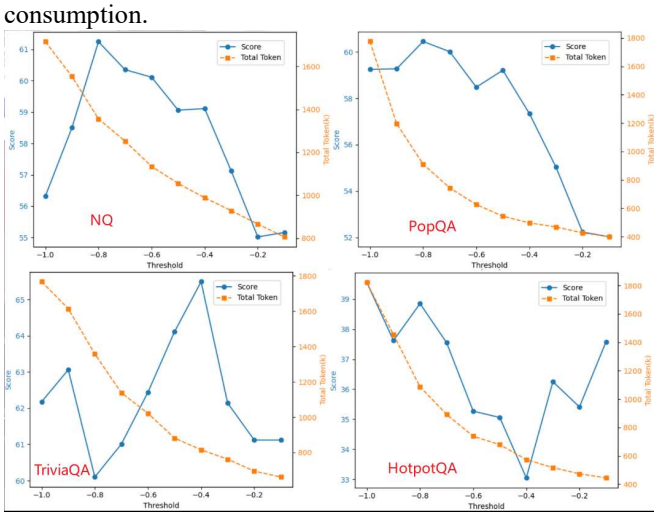

Fig. 5. Balancing token and accuracy score on different datasets.

2) *Ablation Experiment*

To analyze the contributions of fusion module and the filtering module to the performance improvement of TKG-RAG, ablation experiments were conducted on these two modules. Specifically, the augmented fusion was divided into two categories based on the nature of the utilized knowledge: attribute knowledge (ak) and relational knowledge (rk). Additionally, during the information extraction and construction process, topic sentences (ts) for the chunks were generated. An ablation study was also performed to determine whether these topic sentences can enhance overall performance.

Based on the results in TABLE IV. , it is evident that both information fusion and information filtering modules play a crucial role in enhancing overall performance. A detailed analysis reveals that topic sentences (ts) perform best on the PopQA dataset, which despite being a long-tail knowledge dataset, primarily consists of straightforward questions. Conversely, their performance is below on the HotpotQA dataset, likely due to its nature as a multi-hop question-answering dataset, where a single topic sentence may not yield positive effects. The filter was fine-tuned on training sets with relevant data distributions across all datasets, effectively acquiring domain-specific knowledge. Notably, it exhibited the best performance on the NQ dataset, as approximately one-quarter of the questions in the selected NQ test set are related to numerical data.

TABLE IV. ABLATION EXPERIMENTS OF TKG-RAG

Approach	NQ	PopQA	TriviaQA	HotpotQA	LawQA
Naïve-RAG	51.3	55.8	61.5	37.5	58.7
w/o TKG-RAG$_{ak}$	54.5	58.2	64.8	38.3	60.1
w/o TKG-RAG$_{rk}$	56.8	59.2	64.3	36.6	59.5
w/o TKG-RAG$_{ts}$	53.2	58.4	63.5	37.4	60.3
w/o filter	53.1	58.5	63.2	**39.1**	61.2
TKG-RAG	**60.1**	**60.2**	**65.4**	38.2	**62.6**

V. CONCLUSIONS

To utilize the respective advantages of the text-chunk based RAG and the knowledge graph based RAG, we present TKG-RAG, a Retrieval-Augmented Generation framework aimed at enhancing the performance of generation ability of Large Language Models. The framework optimizes the post-retrieval processing stage by making full use of the knowledge about the retrieved text-chunks. Specifically, the text-chunk hierarchy, contextual relationships, and topic sentences in the text-chunked knowledge graph are effectively used to improve the performance of information fusion. In addition, the fine-tuned filtering model can more accurately remove noisy information, leading to a reduction in the number of input tokens required during the generation phase, thereby enhancing the overall performance of Retrieval-Augmented Generation.

Although the experimental results indicate that TKG-RAG can improve the retrieval augmented generation of large language models, several areas can be further investigated in future work. For example, the current information in the text-chunk knowledge graph is insufficient; additional information (such as version details) could be added to enhance the performance of information fusion. Furthermore, TKG-RAG performs poorly on multi-hop question-answering tasks, which could be improved by optimizing question to improve the completeness of retrieval results. In addition, aggregated text-chunks may be mistakenly removed, so an information router could be designed to score questions adaptively to retain potential correct answers. Finally, we aim to explore a broader range of datasets to further validate the generalization ability of TKG-RAG across diverse domains and multilingual settings, and we plan to evaluate it on more challenging benchmark datasets to rigorously assess its robustness.

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